

Recollection of the last SM talk:

- RLD consider duality between pairs

$$(G, M) \longleftrightarrow (\check{G}, \check{M}).$$

where G : (split) red. gp. $\check{G}$: dual

M : hyper spherical variety $\check{M}$: dual hyper spherical variety

hyperspherical includes more cases

$\mathfrak{g}$: Whittaker, theta corresp.

Thm (Structure thm of BZSV).

$$(1) M: G\text{-hypersph. var} \cong (S \times_{\check{H}} S^{\vee})^H \times G.$$

BZSV quadruple (G, H, ι, S)

- (2) If M has a polarization, then:

$$M \cong T^*(X, \Psi) \quad \text{twisted cotangent bundle}$$

$$\text{where } X = S^+ \times^H G \xleftarrow{\text{Ga-torsor}} \Psi = S^+ \times^{H'} G$$

X is a spherical G -variety.

$$T^*(X, \Psi) = T^* \Psi //_1 G_a$$

$$U' = \ker (U \xrightarrow{f} G_m)$$

$$(G, M) \longleftrightarrow (\check{G}, \check{M}) \quad \check{M} \text{ hyperspherical built from } (\check{G}_x, \check{L}_x, S_x)$$

It's too naïve to guess $(G, T^*(H \backslash G))$



$$(G^\vee, T^*(H^\vee \backslash G^\vee))$$

In automorphic point of view. we want to detect duality from period / L-function.

Conj: $(G, M) \longleftrightarrow (G^\vee, M^\vee)$.

$$\begin{aligned} & \pi \in \Pi_2(G) : \text{autom. rep of } G \\ & \downarrow \\ & \psi : L_F \times SL_2 \longrightarrow G^\vee. \end{aligned}$$

$$\frac{|\mathcal{P}_M(f)|^2}{\langle f, f \rangle} \stackrel{''}{=} L_{M^\vee}(\psi) \quad f \in \pi$$

GGP case

Ex, Last time $(SO_{2n} \times SO_{2n-1}, T^*(SO_{2n-1}^\Delta \backslash SO_{2n} \times SO_{2n-1}))$

$$\updownarrow (G, H = SO_{2n-1}, \rho, 1)$$

$$(G, G, \text{std} \otimes \text{std}, 1) \quad (SO_{2n} \times Sp_{2n-2}, V \otimes W)$$

theta case

$$P_M(f) = \int_{[H]} f(h) \Theta_M(h) dh$$

Geometric settings: [GGGPP]

§ Global case: C : curve / $\mathbb{F}_q$ or $\mathbb{C}$. smooth.

Period sheaves & $\mathcal{L}$ -sheaves.

Assume $M = T^*X$ $\check{M} = T^*\check{X}$

$$\begin{array}{ccc} \{(\mathcal{E}, s)\} = \text{Bun}_G^X = \text{Maps}(C, \frac{X}{G}) & & \text{Loc}_{\check{G}}^{\check{X}} = \text{Maps}(C_{\text{dR}}, \frac{\check{X}}{\check{G}}) \\ \downarrow \pi^{\text{aut}} & \text{G-bdle w/ a section} & \downarrow \pi^{\text{Spec}} \\ \{\mathcal{E}\} & \text{of ass. } X \otimes \Omega_C^{\frac{1}{2}}\text{-bdle} & \text{Loc}_{\check{G}} = \text{Maps}(C_{\text{dR}}, B\check{G}) \end{array}$$

Geometric Langlands:

$$\mathcal{D}(\text{Bun}_G) \cong \text{Ind Coh}_{\text{Nilp}}(\text{Loc}_{\check{G}})$$

$$\mathcal{P}_X := \pi^{\text{aut}}_! (\mathbb{C}) \longleftrightarrow \text{Loc}(\pi^{\text{Spec}}_* (\omega))^{\vee} =: \mathcal{L}_{\check{X}}$$

$\left\{ \begin{array}{l} \mathbb{F}_q \text{-case trace of Frob} \end{array} \right.$

$$\mathcal{P}_X: \text{Bun}_G(\mathbb{F}_q) = G(F) \backslash G(A) / G(\hat{O})$$

$$(\mathcal{E}, s), \quad s \in H^0(C, \mathcal{E} \times_{G \times G_m} (X \times \text{Tot}(\Omega_C^{\frac{1}{2}})))$$

$$\Rightarrow \mathcal{P}_X(\mathcal{E}) = \#(H^0(C, \dots))$$

①

or more explicitly:

$$P_X(g) = \sum_{x \in X(F)} (g, \vartheta^{\frac{1}{2}}) \cdot \Phi(x) \quad (g \in G(A))$$

as ① valued at $g \in G(A)$ is the size of

$$X(F) \cap \prod_v X(\mathcal{O}_v) \cdot (g^{-1}, \vartheta^{\frac{1}{2}}) \quad \vartheta^{\frac{1}{2}} = \prod_v \varpi_v^{\frac{n_v}{2}}$$

Φ : characteristic function on $X(\hat{\mathcal{O}})$

$$\mathbb{F}_q: X = G/H \quad \text{Bun}_G^X = \text{Bun}_H \quad \text{over } \mathbb{F}_q$$

$$[H] = \text{Tr}(\text{Frob}_q, \mathcal{P}_X)$$

$$P_H(g_\varepsilon) = \text{Tr}(\text{Frob}_q, \mathcal{P}_X|_\varepsilon) = \# \left\{ \begin{array}{c} H\text{-reduction to } \varepsilon \\ \parallel \\ \text{Bun}_H(\mathbb{F}_q) \end{array} \right\}$$

for $\varepsilon \in \text{Bun}_G(\mathbb{F}_q)$

$$f \in \text{Fun}(\text{Bun}_G(\mathbb{F}_q)) \longleftrightarrow \mathcal{F} \text{ (Weil sheaf)}$$

$$\int_{[G]} P_X(g) f(g) = \sum_{\varepsilon \in \text{Bun}_G(\mathbb{F}_q)} \frac{P_X(g) f(g)}{|\text{Aut}(\varepsilon)|}$$

$$= \int_{\substack{G(A) \\ G(F)}} \sum_{x \in G(F)/H(F)} (g, \vartheta^{\frac{1}{2}}) \Phi(x) f(g)$$

$$= \int_{H(A)} G(A) (g, \partial^{\frac{1}{2}}) \Phi(g) f(g)$$

Eisenstein / Whittaker case. $X = (U, \psi) \backslash G$

$$\Rightarrow \int_{U(F) \backslash G(A)} f(g) (g, \partial^{\frac{1}{2}}) \cdot \Phi(g) \quad \text{(i.e. } X = U \backslash G \text{ w/ } G_A\text{-torsor given by } \psi: U \rightarrow G_A)$$

$\partial^{\frac{1}{2}}$ acting on X by left translation by

$$a_0 := e^{-z\check{\rho}} (\partial^{\frac{1}{2}}) \in T(A_{\mathbb{F}})$$

$$\Rightarrow \text{Supp} (\partial^{\frac{1}{2}} \cdot \Phi) = U(A) a_0 G(\hat{O}).$$

$$\Rightarrow \int_{[G]} P_X(g) f(g) = q^{\beta_X} \int_{[U]} \psi(u) f(u a_0) du$$

Fourier - Whittaker coefficient.

$$\beta_X = \dim \text{Bun}_G^X$$

Next, consider $\mathcal{L}$ -sheaves

$$\text{Recall } \mathcal{L}_{\check{X}} := \left(\pi_*^{\text{Spec}} \omega_{\text{Loc}_{\check{X}}^{\check{G}}} \right)^{\vee} \in \text{IndCoh}(\text{Loc}_{\check{G}})$$

here we view $\pi_* \omega$ as $\bigoplus_n (\pi_* \omega)_n$: weight space of $\mathbb{G}_m$ -action.

$\rightsquigarrow$ shearing $\bigoplus_n (\pi_* \omega)_n \langle n \rangle$

$\rho: \check{G}$ -local system, viewed as $\pi_1(C) \xrightarrow{\rho} \check{G}(\mathbb{k})$

$\rightsquigarrow (l_\rho: \text{pt} \longrightarrow \text{Loc}_{\check{G}}) \in \text{Loc}_{\check{G}}(\mathbb{k})$

$$\delta_\rho := l_\rho * (\mathbb{k}) \quad d = \dim Z(\rho)$$

Notation: $V = \bigoplus V_k \rightsquigarrow L(V^\vee, s) = \prod_k L(s + \frac{k}{2}, V_k)$

$$\pi_1(C) \xrightarrow{(\rho, (\varpi))} \check{G} \times G_m \longrightarrow GL(T)$$

$$T = T_{x_0} X = \bigoplus T_k \quad x_0: \text{isolated } p_{\text{geom}}\text{-fixed point.}$$

$$\rightsquigarrow L(s, T^\vee) = \prod_k L(s + \frac{k}{2}, V_k)$$

$$\bullet \text{Hom}(\mathcal{L}_{\check{X}}, \delta_\rho) = (\text{Sym } H^* T^\vee)^\vee \otimes \det H^1(\text{ad } \rho)$$

$$\Rightarrow [\text{Hom}(\mathcal{L}_{\check{X}}, \delta_\rho)] = q^{-b_G} L(0, T^\vee)$$

$$\bullet \text{Hom}(\delta_\rho, \mathcal{L}_{\check{X}}) = (\text{Sym } H^*(T^\vee))[-d]$$

$$\Rightarrow [\text{Hom}(\delta_\rho, \mathcal{L}_{\check{X}})] = (-1)^d L(1, (T^\vee)^\vee)$$

$H^* T$: cohom. of C w/ coeff. of local sys. defined

by T

Pf: Consider the diagram

$$\begin{array}{ccc} \mathrm{Loc}_\rho^{\check{X}} & \xrightarrow{L_{\check{X}}} & \mathrm{Loc}_{\check{G}}^{\check{X}} \\ \pi_\rho \downarrow & & \downarrow \pi \\ \mathrm{pt} & \xrightarrow{L_\rho} & \mathrm{Loc}_{\check{G}}^{\check{Y}} \end{array}$$

The proof is based
on base-change
in $\mathrm{IndCoh} = \mathrm{QC}!$

$$\begin{aligned} \bullet \quad \mathrm{Hom}(\delta_\rho, L_{\check{X}}) &= \mathrm{Hom}(L_{\rho*} \underline{k}, L_{\check{X}}) \quad \left(\begin{array}{l} d=0 \\ L_\rho \text{ is proper} \end{array} \right) \\ &= L_\rho^! \pi_* \omega_{\mathrm{Loc}_{\check{G}}^{\check{X}}}^{\vee} \stackrel{\parallel}{=} \pi_{\rho*} \omega_{\mathrm{Loc}_\rho^{\check{X}}} \end{aligned}$$

$\mathrm{Loc}_\rho^{\check{X}}$ is just $H^1(T)[-1]$ (derived scheme)

$$\hookrightarrow \text{ring of func} = \mathrm{Sym}(H^1(T)^\vee[1]) \quad \text{assumption here}$$

$$\Rightarrow \text{section of } \omega_{\mathrm{Loc}_\rho^{\check{X}}} = \mathrm{Sym}(H^1(T)[-1]) = \mathrm{Sym}(H^*(T))$$

$d > 0$ case is similar

$$\bullet \quad \mathrm{Hom}(L_{\check{X}}, \delta_\rho) = \mathrm{Hom}(L_\rho^* L_{\check{X}}, \underline{k}) = \mathrm{Hom}(\pi_{\rho*} L_{\check{X}}^* \omega, \underline{k})$$

$$L_{\check{X}}^* = L_{\check{X}}^! \otimes (\det T_\rho)^{-1}$$

$$\Rightarrow L_{\check{X}}^* \omega_{\mathrm{Loc}_{\check{G}}^{\check{X}}} = \omega_{\mathrm{Loc}_\rho^{\check{X}}} (\det T_\rho)^{-1}$$

$$\Rightarrow \text{Hom}(L_{\check{X}}, \delta_\rho) = \left(\text{sec. of } \omega_{\text{Loc}_{\check{X}}^\rho}^\vee \right)^\vee \otimes \det \frac{T_\rho}{H^1(\text{ad } \rho)}$$

Baby case of global geometric conj

= Tate Iwasawa case $X = A^1 \leftarrow G = G_m$

$$Z: \text{Bun}_{G_m} \longrightarrow \text{Bun}_{G_m}^X \quad \text{Zero section}$$

$$j: \text{Bun}_{G_m}^{X,0} \longrightarrow \text{Bun}_{G_m}^X \quad \text{complement.}$$

$$\Rightarrow j_! \underline{\mathbb{k}} \longrightarrow \underline{\mathbb{k}} \longrightarrow Z_* \underline{\mathbb{k}}$$

$$\Rightarrow \pi_!^0 \underline{\mathbb{k}} \longrightarrow \mathcal{P}_X \longrightarrow \underline{\mathbb{k}} \quad \pi^0 = \pi|_{\text{Bun}_{G_m}^{X,0}}$$

$$= \pi_*^0 \underline{\mathbb{k}}[-1]$$

ρ : non-trivial loc. sys. on C .

$\hookrightarrow \mathcal{E}_\rho$: asso. loc. constant sheaf on Bun_{G_m}

s.t. $\mathcal{E}_\rho$ pull-backs to $\rho^{\boxtimes r}$ under

$$C^r \longrightarrow \text{Bun}_{G_m}$$

$$(P_1, \dots, P_r) \longmapsto \mathcal{O}(\sum P_i)$$

$$\mathcal{F}_\rho := \mathcal{E}_\rho \langle g-1 \rangle \longleftrightarrow \delta_\rho \in \text{QC}^!(\text{Loc}_{G_m})$$

$$\begin{aligned}
\mathrm{Hom}(\mathcal{F}_\rho, \mathcal{P}_X \langle g-1 \rangle) &= \mathrm{Hom}(\mathcal{E}_\rho, \mathcal{P}_X) \\
&= \mathrm{Hom}(\mathcal{E}_\rho, \pi_\star^0 \underline{k}[-1]) = \mathrm{Hom}((\pi^0)^* \mathcal{E}_\rho, \underline{k})[-1] \\
&= \mathrm{Sym} H^*(\Sigma, \check{\rho})[-1] \cong \mathrm{Hom}(\delta_\rho, \mathcal{L}_{\check{\rho}}).
\end{aligned}$$

$\hookrightarrow$ dual local system.

Local case:

Recall the (geometric) Satake:

$$(\text{Perv}(G_0 \backslash G_r), *) \cong (\text{Rep}(\check{G}), \otimes).$$

The derived Satake (Bez-Fink, Ginz):

Spherical
Hecke
cat.

$$\begin{aligned} (\text{Shv}(G_0 \backslash G_r), *) &\cong (\text{Perf}(k[\check{\mathfrak{g}}^*]^{\vee}))^{\check{G}^{\vee}}, \otimes \\ (\text{SHV}(G_0 \backslash G_r), *) &\cong \text{QC}((\check{\mathfrak{g}}^*/\check{G})^{\vee}) \end{aligned}$$

$$(G, M = T^*(X, \Psi)) \quad (\check{G}, \check{M})$$

$\rightsquigarrow$ (unramified) local geometric conjecture.

$$\mathbb{L}_X: \text{SHV}(X_F/G_0) \cong \text{QC}^{\vee}(\check{M}/\check{G})$$

$$\bullet \mu: \check{M}/\check{G} \longrightarrow \check{\mathfrak{g}}^*/\check{G} \text{ moment map is } G_{\text{gr}}\text{-equiv}$$

$$\Rightarrow \text{QC}^{\vee}(\check{M}/\check{G}) \text{ is a module category for } (\text{QC}^{\vee}(\check{\mathfrak{g}}^*/\check{G})).$$

$$\bullet \text{Affineness of } \check{M} \rightsquigarrow \check{M}/\check{G} \text{ affine over } \check{\mathfrak{g}}^*/\check{G}$$

$$\Rightarrow \text{QC}(\check{M}/\check{G}) = \mu_* \mathcal{O}\text{-mod}$$

$$\mu_* \mathcal{O}: \text{algebra obj in } \text{QC}(\check{\mathfrak{g}}^*/\check{G})$$

$$\Rightarrow QC^{\mathbb{I}}(\check{M}/\check{G}) \cong \mu_* \mathcal{O}^{\mathbb{I}} - \text{mod}_{QC^{\mathbb{I}}(\check{A}^*/\check{G})}$$

• Basic obj in $SHV(X_F/G_0)$ is

$$\delta_X = i_* \omega_{X_0} \quad i: X_0/G_0 \longrightarrow X_F/G_0$$

$SHV(X_F/G_0)$ naturally carries an Hecke act^o

$$\text{Conj: } \mathbb{L}_X(\delta_X) \cong \mathcal{O}_{\check{M}/\check{G}}^{\mathbb{I}} \quad PL_X := \text{End}_{\overline{\mathcal{H}}_G}(\delta_X)$$

we should have: $\in \text{Alg}(\overline{\mathcal{H}}_G)$

$$\bullet \quad k[\check{M}/\check{G}]^{\mathbb{I}} = (k[\check{M}]^{\check{G}})^{\mathbb{I}} \cong \text{End}(\delta_X) = H_G^*(X)$$

$$\bullet \quad \text{Hom}(\tau_V * \delta_X, \tau_W * \delta_X) \cong \text{Hom}^{\check{G}}(V, W \otimes \mathcal{O}_{\check{M}}^{\mathbb{I}})$$

$$\text{Conj: } PL_X \cong \mathcal{O}(\check{M})^{\mathbb{I}}$$

What we get if we put $[\dots]$ on both sides? Plancherel formula.

$$\text{Eg: } X=H \curvearrowright G=H \times H$$

$$\mathcal{H}_G = \mathcal{H}_H \otimes \mathcal{H}_H \curvearrowright S(X(F))^{G(0)} \cong \mathbb{C}[\check{A}]^W \cong \mathbb{C}[\text{Rep } \check{H}]$$

we have the Plancherel formula:

$$\langle h_1, h_2 \rangle = \frac{1}{|W|} \int_{\check{A}} \frac{\hat{h}_1(t) \hat{h}_2(t^{-1})}{\prod_{\alpha \in \Phi} (1 - q^{-1} e^{\check{\alpha}})} d_{Weyl} t \frac{1}{(1 - q^{-1})^{rk H}}$$

$$= \frac{1}{|W|} \int_{\check{A}} \hat{h}_1(t) \hat{h}_2(t^{-1}) L(t, \check{h}, 1) d_{Weyl} t$$

Take $h_2 = \mathbb{1}_{H(0)}$

$$= \frac{1}{|W|} \int_{\check{A}} \hat{h}_1(t) \sum_{i \geq 0} q^{-i} \text{tr}(t | S^i \check{h}^\vee) d_w t.$$

Take $\hat{h}_1 = h_V \quad V \in \text{Im}_{\check{H}}$

$$= \sum_{i \geq 0} q^{-i} \dim \text{Hom}(V, S^i \check{h}^\vee)$$

$$= \left[\text{Hom}(V, \mathbb{K}[\check{h}^*]^{\vee}) \right] = \langle \text{tr Frob } \mathbb{I}_{C_V}, \text{tr Frob } \mathbb{K}_{H_0} \rangle$$

Now since $\check{M} = V_X \overset{G_X^\vee}{\times} \check{G}$

we want $\langle h * \mathbb{1}_{X_0}, \mathbb{1}_{X_0} \rangle$

$$= \int_{\check{G}_X} \hat{h}(t) L(t, V_X, 1) d_w t$$

$\hat{h}$ for Satake transform

Some computations of δ_X :

$$X = \mathbb{A}^1 \curvearrowright G = \mathbb{G}_m \quad (\text{Iwasawa - Tate})$$

$$X_F = \varinjlim X^\ell \quad X^\ell = \varprojlim_n X_n^\ell$$

$$X^\ell(R) = t^{-\ell} R[t] \quad X_n^\ell(R) = t^{-\ell} R[t]/t^n$$

$$\delta_\ell := \underline{k}_{X^\ell} \langle \ell \rangle \rightsquigarrow [\delta_\ell] = q^{-\frac{\ell}{2}} \mathbb{1}_{t^{-\ell} X_0}$$

$$\mathcal{T}_{V_\ell} * \delta_X = \delta_\ell \quad \forall \ell \in \text{Rep}(\mathbb{G}_m = \check{G}): \mathbb{Z} \mapsto \mathbb{Z}^\ell$$

$$\cdot \underline{k}_{X^\ell} \xrightarrow{\textcircled{1}} \underline{k}_{X^{\ell-1}} \quad ; \quad \underline{k}_{X^\ell} \xrightarrow{\textcircled{2}} \underline{k}_{X^{\ell+1}} \langle 2 \rangle$$

(comes from inclusion of
dualizing sheaf = trivial ones)

$$\delta^\circ = \text{colim}_{\textcircled{1}} \underline{k}_{X^\ell}$$

$$\mu = \text{colim}_{\textcircled{2}} \underline{k}_{X^\ell}$$

$$\Rightarrow [\text{Hom}(\mathcal{F}, \delta^\circ)^\vee] = f(0) \mu(X_0/G_0)$$

$$[\text{Hom}(\mathcal{F}, \mu)^\vee] = \int_{X_F/G_0} f(x) dx$$

$\mathcal{F}$: Weil sheaf w/ corresp function f .